

Flyhomes BBYS Mortgage Calculator: Master Project Bible

1. Project Overview & Persona

A "Buy Before You Sell" (BBYS) mortgage calculator for Flyhomes. Focus on unlocking equity and providing smart, scenario-based loan recommendations.

Team Persona

- **UX Designer:** Focus on "finance-grade" UI, high-trust, and clear benefit visualization.
 - **Loan Officer:** Ensure 100% math accuracy and precise lending terminology.
 - **Lead Dev:** Expert in clean, single-file HTML/Tailwind/Vanilla JS.
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2. 🚀 Workflow Orchestration (Design-Driven Development)

Plan Before Building

- **Visual Plan First:** For any UI task, output a brief plan (layout, components, spacing) **before** writing code.
- **Ask, Don't Guess:** If direction is unclear, ask the designer first. Stop and re-align immediately when blocked.

Handle Token Limits Proactively

- **Phase-Based Work:** Break complex tasks into: Structure -> Logic/Math -> Interactions/UI.
- **No Over-Output:** Summarize at the end of each phase. Never try to complete all code in a single output.

Simplicity Over Cleverness

- **Direct Approach:** Implement in the simplest way possible; avoid over-abstraction.
- **Readability:** Code readability matters more than complexity. If a solution feels convoluted, stop and ask.

Design Consistency & Iteration

- **Reuse Tokens:** Strictly reuse existing colors/spacing. Don't add new ones arbitrarily.
- **Iteration-Friendly:** Only touch what needs changing—don't rewrite entire files for small tweaks.
- **Intent Over Literal:** Understand the *design intent* behind visual feedback—don't just literally translate words into code.

Visual Verification

- **Self-Check:** Before delivering, verify visual hierarchy, alignment, and responsiveness.

- **Explicit Signal:** Tell the user "Ready to preview in browser"—do not silently finish.
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3. Branding & Design System

- **Primary Color:** #4C7994 (Flyhomes Blue).
 - **Primary CTA:** Background #D9848B (pink), text #232226 (near-black). Hover #CB7178 . Apply to ALL primary CTAs (e.g., nav "Get Started", "Submit scenario to get official offer"). Do not use the brand blue for primary CTAs.
 - **Top Nav:** Dark background #1a1a1a , white Flyhomes wordmark logo, gray-200 link text, hover white.
 - **Typography:** Libre Franklin (Google Fonts, weights 300/400/500/600/700) is the project font. Do not use Inter.
 - **Tokens:** Success (#10B981), Warning (#F59E0B), Border (#E5E7EB).
 - **Style:** Ant Design patterns, generous whitespace, high-trust minimalist UI.
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4. DREAM Solutions & Recommendation Engine

A. Challenge Selection (The 4-Option UI)

Confirmed 2026-08-14: card copy below updated per compliance/copy spec

(BBYS_Calculator_Copy_Spec_8_13.pdf) — global no-dash rule (no em dashes, en dashes, or hyphens in visible copy), plain buyer-first pain framing. Solutions/rules unaffected.

1. **DTI Issue:** "My current mortgage is capping my approval"
 - *Subtitle:* You cannot carry two mortgages at once, so your approval on the new home comes in lower than you need.
2. **Liquidity Issue:** "My cash is locked in my current home"
 - *Subtitle:* Your down payment and closing costs are sitting in your current home, stuck until it sells.
3. **Strategy/Speed:** "My offers keep losing to cash buyers"
 - *Subtitle:* You wish your offer could be as strong as cash, winning without having to bid the highest. (Also supports mortgage-free downsizing.)
4. **Retirement Focus:** "I want to downsize, ideally without a new mortgage"
 - *Subtitle:* You have plenty of home equity but not enough income to qualify for a new loan the usual way.

B. Trigger Rules (Solution Mapping)

- **Rule 0: The Downsize Override (Priority)**
 - **Condition:** IF (Option 4 IS checked)
 - **Result:** Show **Retire & Downsize** solution ONLY. This selection overrides all other logic combinations — except Rule 5 (Cross-State Block), which still applies to downsize since Retire & Downsize is a Cross Collateral product and CC requires same-state homes. See Rule 5.
- **Rule 1: The "Cross Collateral" Winner**
 - **Condition:** IF (Option 2 IS checked AND Option 3 IS checked)

- **Result:** Show **Cross Collateral** ONLY. (Hide IE/CO/GBC cards as CC is the integrated \$0 cash/downsize king).
- **Rule 2: Equity Focused**
 - **Condition:** IF (Option 2 checked ONLY) OR (Option 1 + 2 checked)
 - **Result:** Show **Equity for Down Payment** (IE) + **DTI Buster** (GBC).
- **Rule 3: Cash Offer Focused (Auto-Gating with Toggle Override)**
 - **Condition:** IF (Option 3 checked ONLY) OR (Option 1 + 3 checked)
 - **Evaluation:**
 - Check the state of the "Keep More Cash" override flag (`S.forceCashOffer`).
 - Calculate `projectedFinalLTV = (NewHomePrice + CurrentMortgage - (DepartingPrice * 0.9)) / NewHomePrice` .
 - **Result:**
 - IF `S.forceCashOffer` is TRUE: Show **All-Cash Advantage** (CO) + **DTI Buster** (GBC) immediately.
 - IF `S.forceCashOffer` is FALSE AND `projectedFinalLTV <= 0.75` : Show **Cross Collateral** ONLY.
 - IF `S.forceCashOffer` is FALSE AND `projectedFinalLTV > 0.75` : Show **All-Cash Advantage** (CO) + **DTI Buster** (GBC).
- **Rule 5: Cross-State Block**
 - **Condition:** IF the user selects different states for "What state is your current home in?" and "What state will you be buying in?" (both fields populated, values differ)
 - **Result:** Do NOT recommend **Cross Collateral** regardless of any other rule, **including Rule 0 (Downsize Override)**. Cross Collateral requires both homes to be in the same state — no exception for downsize.
 - **Rule 0 downsize blocked:** Fall back to **Instant Equity** (IE) + **DTI Buster** (GBC). Downsize alone (Option 4, no Option 3) carries no competitive-cash-offer intent — its core value prop is "use equity toward the new home," which Instant Equity preserves. Cash Offer does not fit here since it solves a different problem (winning a bidding war) that a pure-downsize user never asked for.
 - **Rule 1 equity+cash combo blocked:** Fall back to **All-Cash Advantage** (CO) + **DTI Buster** (GBC), since the user explicitly selected Option 3 (competitive cash offer).
 - Fall back to **DTI Buster** (GBC) alone for DTI-only intent.
 - **Confirmed 2026-08-13** (fixes §6.1 bug): when Rule 5 blocks a downsize-triggered CC, the UI must also switch its title/badge/subtitle away from "Retire & Downsize"/"Cross Collateral" to match whatever product actually gets computed — the label must always be derived from the resolved product, never from the raw Option 4 checkbox alone. Do not maintain the solution-card label in two separate places (e.g. once from the resolved calc() branch, once again from raw checkbox state) — that duplication is what let the label and the math disagree.
- **Rule 4: Pure DTI**
 - **Condition:** IF (Option 1 checked ONLY)
 - **Result:** Show **DTI Buster** (GBC) only.

C. UI Badges & Labels

- **"Most Likely Needed":** Apply this badge to the **DTI Buster** card if Option 1 was **NOT** manually checked but Rule 2 or Rule 3 triggered it.
- **Downsize Hint:** If Option 3 is active, ensure UI emphasizes: "Perfect for mortgage-free downsizing."

D. Scenario Branding: "Retire & Downsize"

- **Trigger:** Option 4 selected.
 - **UI Title:** "Retire & Downsize"
 - **Product Badge:** "Cross Collateral"
 - **Main Subtitle:** "Leverage home equity to downsize for retirement and enjoy greater financial freedom, often without taking on a new long-term mortgage."
 - **Logic Sync:** This is a marketing skin for the **Cross Collateral** product. It must use all formulas and data points defined for Cross Collateral in Section 5.D.
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5. Calculation Logic (Source of Truth)

A. Total Estimated Upfront Cost Formula

Total Estimated Cost = (Loan Amount × Origination Fee %) + GBC Fee + Accrued Interest

Accrued Interest

- Rate: 9.99% per annum
- Formula: **Loan Amount × 9.99% × (transitionDays / 365)**
- Only shown when **Loan Amount > 0** (hidden in DTI-only scenario where there is no loan).
- Displayed as a separate line item in the Estimated Cost dropdown alongside "Other Costs".

B. Origination Fee % Rules

- **BBYS + Cash Offer & Cross Collateral:**
 - If **LTV > 90.00%**: 1.5%
 - If **LTV ≤ 90.00%**: 1.0%
- **Instant Equity:**
 - If **1st Lien**: 2.0% | **2nd Lien**: 2.5%
 - **Confirmed 2026-08-13:** this calculator is a preliminary/directional estimate, not a bindable quote, so lien-position detection is intentionally NOT implemented. The app has no lien-position input and none of the existing fields (e.g. **Current Mortgage Balance**) reliably imply it — whether an existing mortgage forces 2nd lien or gets paid off/subordinated to keep the new loan in 1st position depends on the specific lending product structure, which this app doesn't model. **originationRate** is hardcoded to **0.02** (1st lien) everywhere (index.html:1245). Do not "fix" this by inferring lien position from **Current Mortgage Balance** without a new product decision — that was explicitly considered and rejected as unreliable.

C. GBC Fee Matrix (Standalone vs. Bundle)

Determined by the **Departing Home Price**. Standalone applies if no Flyhomes loan is selected. *Note: Cross Collateral CANNOT be combined with GBC Fee.*

Formula: **Net Bundle Fee = Standalone Fee – Bundle Credit**

Guaranteed Price Range	Standalone Fee (No Loan)	Bundle Credit	Net Bundle Fee (With Loan)
Up to \$500,000	\$2,500	\$0	\$2,500
\$500,001 – \$750,000	\$3,500	\$1,000	\$2,500
\$750,001 – \$1,000,000	\$5,000	\$2,500	\$2,500
\$1,000,001 – \$1,500,000	\$7,500	\$2,500	\$5,000
\$1,500,001 – \$2,000,000	\$10,000	\$2,500	\$7,500
Above \$2,000,000	Exception	Exception	\$10,000+ (exception — requires manual handling; not auto-calculated)

Exception rule for Above \$2,000,000: Both the Standalone Fee and Bundle Credit fall outside the standard tier table. The system must flag this as an exception. Default display value is \$10,000+, but the actual fee must be confirmed manually and is not programmatically computed.

Confirmed 2026-08-13 — input is capped at \$2,000,000, not left open with an exception state: "Estimated Value of Your Current Home" (`S.homeValue` , index.html:319-326) has its slider max set to `2,000,000` (not \$3M). If the user types a value above \$2,000,000 directly into the text field, `render()` (index.html:1543) hides the entire `#results-panel` and shows "Please contact us to learn more." under the field instead of computing an estimate. This avoids ever hitting the >\$2M GBC exception path in a way that silently produces an incomplete total (see §6.5-style gap) — above \$2M, the tool simply doesn't attempt an estimate.

D. LO Broker Fee Rules (Estimated Average)

Programmatically derive the Broker Fee based on the active product track and its corresponding loan/advance base amount:

- **BBYS + Cash Offer Track:** `BrokerFee = FCO_LoanAmount * 0.01` (1% of the standalone Cash Offer loan amount)
- **Instant Equity Track:** `BrokerFee = IE_AdvanceAmount * 0.005` (0.5% of the unlocked equity advance amount)
- **Cross Collateral Track:** `BrokerFee = CC_CappedLoanAmount * 0.01` (1% of the final dual-capped Cross Collateral loan amount)

UI & Multi-panel Guidelines:

1. **Encapsulation:** These calculated fees must not be shown as a standalone raw line item to the buyer. They must be rolled into the `Other Costs` parent element.
2. **Tooltip Rendering:** Inside the `Other Costs` disclosure tooltip, map this value to the professional label: `Loan Processing & Underwriting: $[Calculated_Value]` .
3. **Reactivity:** Any change to input variables (e.g., purchase price or departing home value) that shifts the underlying loan amount must instantly re-run these percentage models to keep the totals synchronized.

E. Core Product Formulas

Instant Equity: $(\text{Departing Price} * 0.75 * 0.9) - \text{Current Mortgage}$. Max LTV: 90% of GBC Price.

Max Purchase Price: $(\text{Liquid Assets} + \text{Instant Equity}) / 0.05$. Ensures 5% Down Payment can be funded by both Cash and Home Equity.

BBYS + Cash Offer: $\text{New Home Price} * 95\%$. Max LTV: 95%. Min Down Payment: 5%.

Cross Collateral: $\text{MIN}((\text{New Home Price} * 1.05), ((\text{New Home Price} + \text{Departing Price}) * 0.75 - \text{Current Mortgage}))$.

Hard risk caps: 105% Acquiring LTV maximum and 75% Combined CLTV maximum.

F. Estimated Savings Logic

Trigger: Display in "Review Results" section to replace the old "Moving Once" module.

Data Provenance:

departingHomePrice: Live-synced from Step 2 "Estimated Departing Home Value" input/slider.

newHomePrice: Live-synced from Step 2 "Estimated Purchase Price" input/slider.

isCashOfferSelected: Boolean flag from Step 1 selection.

1. Cash Offer Price Advantage (Conditional) Visibility: Show whenever the recommended product uses a cash offer — i.e. `S.cash` (BBYS + Cash Offer track) OR `S.noCash` (Cross Collateral, including Retire & Downsize) is true. Do NOT gate this further on a standalone liquid-asset check (e.g. `assets >= newPrice * 0.05`) — cash-offer eligibility already accounts for combined cash + equity (`combinedResources`) elsewhere in the app, and re-checking liquid assets alone here can hide the benefit from a user who was legitimately recommended the Cash Offer product. (Confirmed 2026-08-13, fixing a prior undocumented gap where this exact liquid-asset re-check suppressed the row.)

Logic: Default 3.5% (Slider: 1% - 10%).

Formula: `newHomePrice * cashOfferDiscount%`

Tooltip: "A cash-strong offer can win at a lower price than a financed one. Drag the slider to set how big a discount to estimate; our 3.5% default is the conservative floor."

2. Staged Home Premium Logic: Default 5% (Slider: 1% - 10%).

Formula: `departingHomePrice * stagedHomePremium%`

Tooltip: "Staging an empty home can lift the sale price. This shows the estimated increase minus a typical \$3,500 staging cost."

3. Home Price Appreciation (HPA) Logic: Default 4% (Slider: -10% - 10%).

Formula: `newHomePrice * (hpaRate% / 12) * (transitionDays / 30)`

Note: Buy-side only — counts only the price increase avoided on the new purchase. The sell-side gain is excluded. Driven by the user-editable `transitionDays` field (default 60 days).

Tooltip: "Based on the annual price-change rate you set. This counts only the buy-side savings from acting now — we always recommend selling your current home as soon as possible."

4. "Move Once, Not Twice" (Logistics) Layout: Single accordion item containing three editable text fields.

Tooltip: "Estimated savings are based on the extra cost of moving twice and two months of transition, compared with moving only once."

Sub-Items (Auto-calculated but editable):

Extra Moving Fee: $(\text{departingHomePrice} * 0.003) + 500$

Temporary Housing: $(\text{departingHomePrice} * 0.006) * (\text{transitionDays} / 30)$

Storage Fee: $(150 + \text{departingHomePrice} * 0.0002) * (\text{transitionDays} / 30)$

Category Total: Sum of the three sub-fields above.

transitionDays: User-editable input field in the results panel (above Estimated Cost). Default: 60. Minimum: 1. Affects HPA, Temporary Housing, and Storage Fee calculations.

5. Grand Total Calculation: Sum of all active items (Note: Cash Offer Discount = 0 if hidden).

UI: Display "Total Estimated Savings" prominently at the bottom of the section.

6. Technical Implementation

- **Single File:** All CSS (Tailwind), HTML, and JS in `index.html`.
- **Reactivity:** Use a single state object `S`. Any input change triggers `calc()` -> `render()`.
- **UX:** Currency formatting on blur; bidirectional slider sync; keyboard-accessible tooltips.

7. Capture & Learn

- **Self-Correction:** After any correction on visuals or structure, **update this file** with the new rule.
- **Goal:** The same class of mistake must not repeat.

This is a point-in-time export of the project's living design/logic doc (`CLAUDE.md`). The source file may have changed since export — treat the repo version as authoritative.